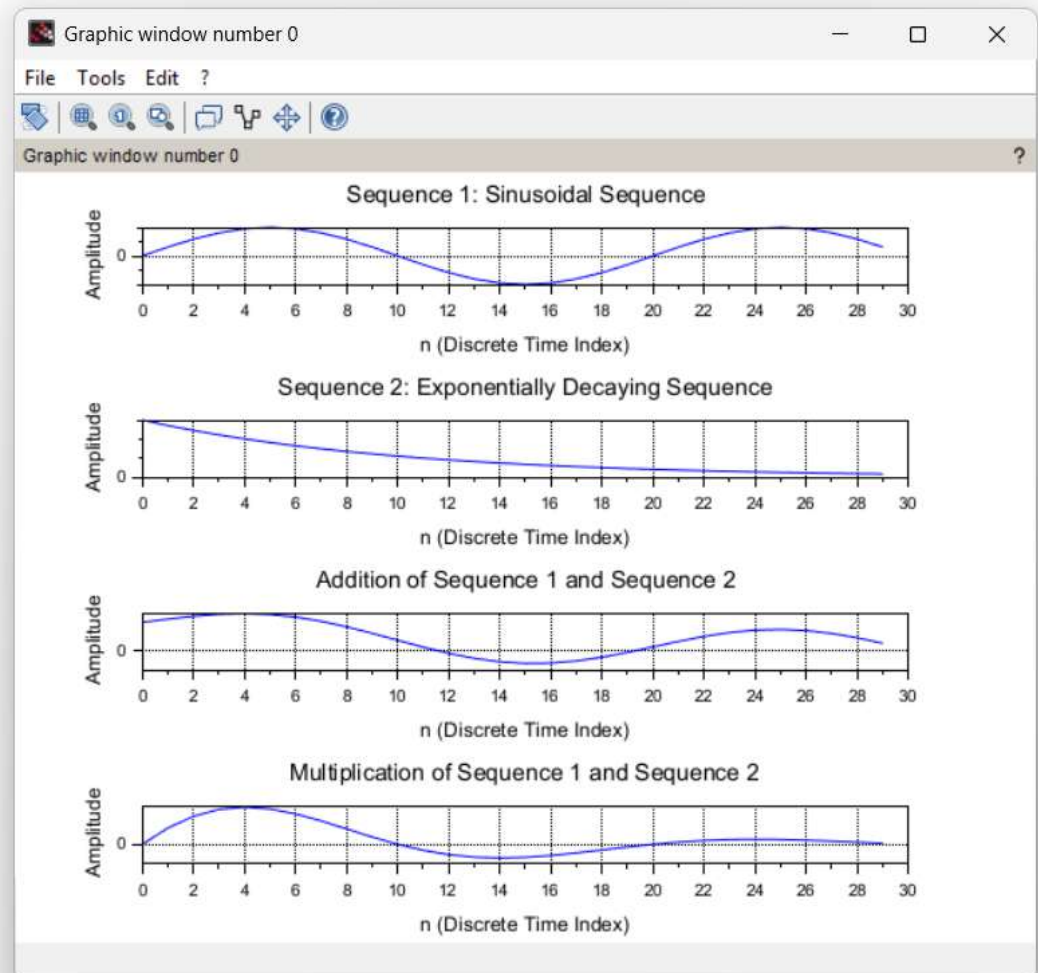




EXP_3.1 SS NEW.sci (E:\EXP_3.1 SS NEW.sci) - SciNotes

EXP_3.1 SS NEW.sc

```
15 a = 0.1;  
16 A2 = 3;  
17 x2 = A2 .* exp(-a .* n);  
18  
19 // Addition  
20 x_add = x1 + x2;  
21  
22 // Multiplication  
23 x_mul = x1 .* x2;  
24  
25 // Plot Sequence 1  
26 subplot(4,1,1);  
27 plot(n, x1);  
28 xlabel("n (Discrete-Time-Index)");  
29 ylabel("Amplitude");  
30 title("Sequence-1: Sinusoidal-Sequence");  
31 xgrid();  
32  
33 // Plot Sequence 2  
34 subplot(4,1,2);  
35 plot(n, x2);  
36 xlabel("n (Discrete-Time-Index)");  
37 ylabel("Amplitude");  
38 title("Sequence-2: Exponentially-Decaying-Sequence");  
39 xgrid();  
40  
41 // Plot Addition  
42 subplot(4,1,3);  
43 plot(n, x_add);  
44 xlabel("n (Discrete-Time-Index)");  
45 ylabel("Amplitude");  
46 title("Addition-of-Sequence-1-and-Sequence-2");  
47 xgrid();  
48  
49 // Plot Multiplication  
50 subplot(4,1,4);  
51 plot(n, x_mul);  
52 xlabel("n (Discrete-Time-Index)");  
53 ylabel("Amplitude");  
54 title("Multiplication-of-Sequence-1-and-Sequence-2");  
55 xgrid();  
56
```



```

1 //Test Case: Autocorrelation of a Rectangular Pulse Sequence
2 //Define parameters
3 N = 20; //Number of samples
4 x_n = [ones(1,5), zeros(1,N-5)]; // Rectangular pulse sequence with width 5
5 // Compute the autocorrelation
6 R_xx = corral(x_n, x_n);
7 // Create a lag vector for plotting
8 lags = -N+1:N-1;
9 // Plot the input sequence
10 clf; // Clear the current figure
11 subplot(2,1,1); bar(0:N-1, x_n, 'b');
12 xlabel('n (Discrete Time Index)');
13 ylabel('Amplitude');
14 title('Input Sequence: Rectangular Pulse Sequence');
15 // Plot the autocorrelation sequence
16 subplot(2,1,2); bar(lags, R_xx, 'r');
17 xlabel('Lag');
18 ylabel('Autocorrelation');
19 title('Autocorrelation of the Rectangular Pulse Sequence');
20

```

